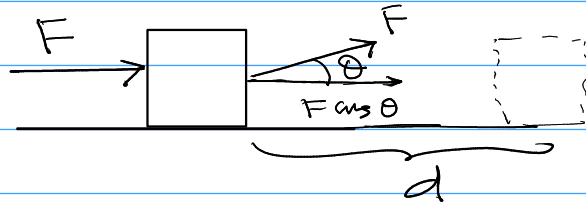


Work and Energy

Now:



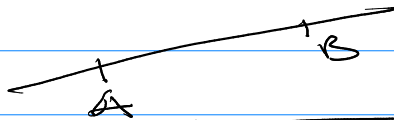
$$W = \vec{F} \cdot \vec{d} = F d \cos \theta$$

$$W = F \cos \theta \cdot d \cos 0^\circ$$

$$W = F d \cos \theta \quad \checkmark$$

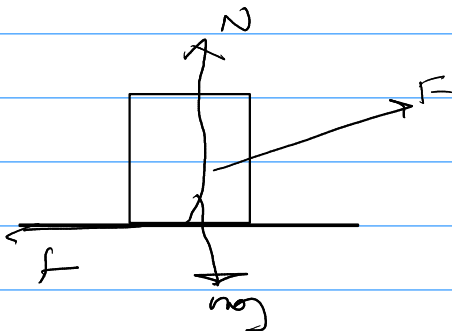
$$\vec{F} = -kz \hat{z}$$

$$F_s = -kz \quad \uparrow$$



$$W = \int_A^B F(\vec{r}) d\vec{r} \quad \checkmark$$

$$W = \begin{cases} > 0 \\ = 0 \\ < 0 \end{cases}$$



$$W_{\text{tot}} = W_n + W_g + W_a + W_f$$

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$W = \vec{F} \cdot d\vec{r} = F_x dx + F_y dy + F_z dz$$

$$W = \int_{\vec{r}_i}^{\vec{r}_f} \vec{F}(\vec{r}) \cdot d\vec{r} = \int_{x_i}^{x_f} F_x dx + \int_{y_i}^{y_f} F_y dy + \int_{z_i}^{z_f} F_z dz$$

$$W_a = \vec{F} \cdot \vec{d}$$

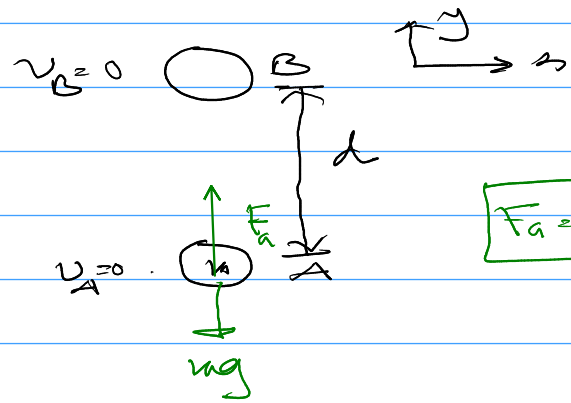
$$W_a = F d \cos \hat{r}$$

$$\Rightarrow W_a = F d$$

$$\Rightarrow \boxed{W_a = +mg d}$$

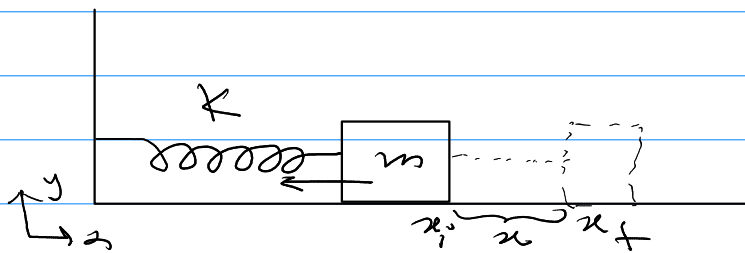
$$W_g = \vec{F}_g \cdot \vec{A} = mg d \cos 180^\circ$$

$$\Rightarrow \boxed{W_g = -mg d}$$



$$W_{\text{net}} = W_a + W_g$$

$$\Rightarrow \boxed{W_{\text{net}} = 0}$$



$$\boxed{x_f - x_i = x}$$

$$W = \int_{x_i}^{x_f} F(x) dx$$

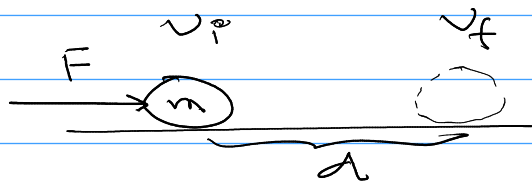
$$W = \int_{x_i}^{x_f} -kx dx = -k \int_{x_i}^{x_f} x dx$$

$$W = -k \left[\frac{x^2}{2} \right]_{x_i}^{x_f}$$

$$\Rightarrow \boxed{W = \frac{1}{2} k x_i^2 - \frac{1}{2} k x_f^2}$$

$$F = -kx$$

Work energy theorem:



$$\vec{F} = m\vec{a}$$

$$a = \frac{F}{m}$$

$$v_f^2 = v_i^2 + 2ad$$

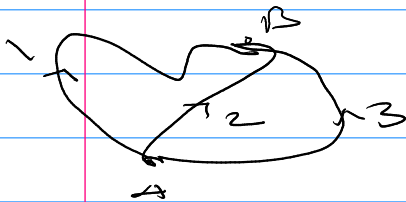
$$\Rightarrow v_f^2 = v_i^2 + 2 \frac{F}{m} d$$

$$\Rightarrow \frac{m}{2} v_f^2 = \frac{m}{2} v_i^2 + \frac{m}{2} 2 \frac{F}{m} d$$

$$\Rightarrow \frac{1}{2} m v_f^2 = \frac{1}{2} m v_i^2 + Fd$$

$$\Rightarrow K_f = K_i + W$$

$$\Rightarrow \boxed{W_{\text{net}} = K_f - K_i = \Delta K}$$



$$\boxed{W_1 = W_2 = W_3}$$

$$\boxed{W_1 \neq W_2 \neq W_3}$$

$$\boxed{W_{\text{net}} = \Delta K}$$

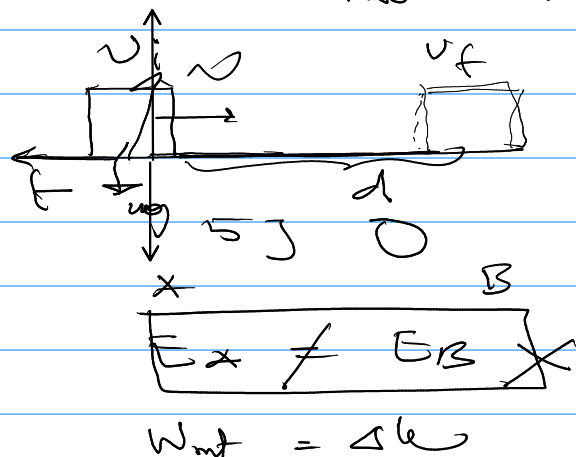
$E_i = E_f$
conservation of
mechanical energy

$$W_N = 0$$

$$W_g = 0$$

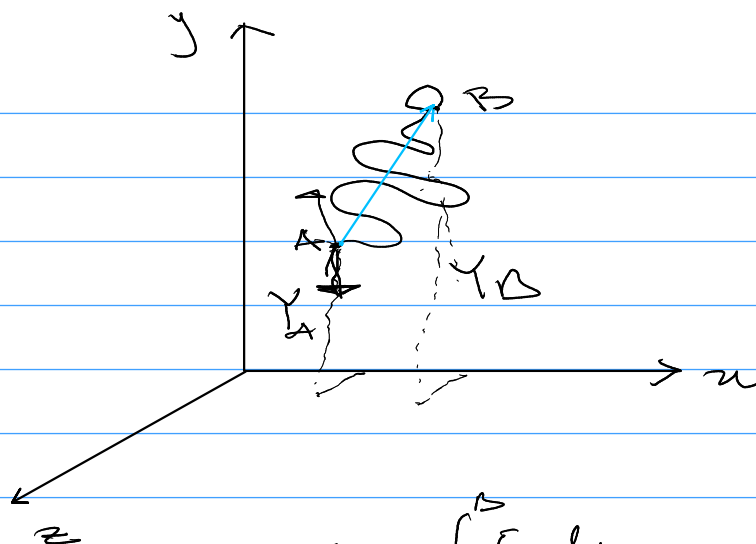
$$\boxed{W_f = -fud}$$

$$\boxed{E_A + W_f = E_B}$$



$W_{\text{net}} > 0, \quad v_f > v_i$
 $< 0, \quad v_f < v_i$
 $= 0, \quad v_f = v_i$

constant acceleration
constant velocity



$$y_B - y_A = h$$

$$W_g = \int_A^B \vec{F}_g \cdot d\vec{y}$$

$$W_g = - \int_A^B mg \, dy$$

$$\Rightarrow W_g = -mg \int_A^B dy = -mg [y]_A^B$$

$$\Rightarrow W_g = -mg [y_B - y_A]$$

$$\Rightarrow W_g = -mgh$$

$$U = mgh$$

$$\Rightarrow \Delta W_c = -\Delta U$$



$$\Delta W_c = -\Delta U$$

$$\Delta W_{nc} = \Delta E$$

$$E_i \neq E_f$$

$$E_i + W_f = E_f$$

$$\Rightarrow W_f = E_f - E_i$$

$$\Rightarrow \Delta W_{nc} = \Delta E$$

$$W_c = -\Delta U$$

$$\Delta W_{nc} = \Delta E$$

If system is conservative

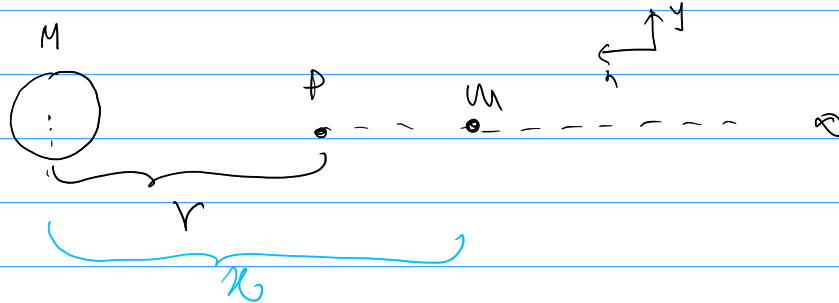
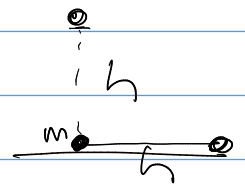
$$\Delta E = 0$$

$$E_i = E_f$$

If system is nonconservative, $\Delta E = W_{nc} \Rightarrow E_f - E_i = \Delta E \Rightarrow \Delta E \neq 0$

Gravitational Potential:-

$$U = mgh$$



$$F = \frac{GMm}{r^2}$$

$$W = \int_{\infty}^r \vec{F} \cdot d\vec{r}$$

$$W = \int \vec{F} \cdot d\vec{r}$$

$$W = \int_{\infty}^r F dr$$

$$W = \int_{\infty}^r \frac{GMm}{r^2} dr$$

$$\Rightarrow W = GMm \int_{\infty}^r \frac{1}{r^2} dr$$

$$\Rightarrow W = GMm \left[-\frac{1}{r} \right]_{\infty}^r$$

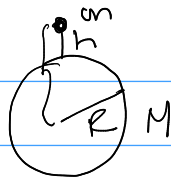
$$\Rightarrow W = -GMm \left[\frac{1}{r} - \frac{1}{\infty} \right]$$

$$\Rightarrow W = -\frac{GMm}{r} = U$$

$$\Rightarrow W = U = -\frac{GMm}{r}$$

$$U = -\frac{GMm}{r}$$

$$U = mgh$$



$$U = -\frac{GMm}{(R+h)}$$

$$\approx U_2 = -\frac{GMm}{R(1+h/R)}$$

$$\approx U_2 = -\frac{GMm}{R} \left(1 + \frac{h}{R}\right)^{-1}$$

$$\approx U_2 = -\frac{GMm}{R} \left(1 - \frac{h}{R}\right)$$

$$\approx U_2 = -\frac{GMm}{R} + \left(\frac{GMmh}{R^2}\right)$$

$$\approx U_2 = -\frac{GMm}{R} + mgh$$

$$\approx U_2 = mgh$$

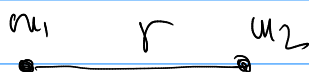
$$U_2 = -\frac{GMm}{R}$$

$$F = ma = mg$$

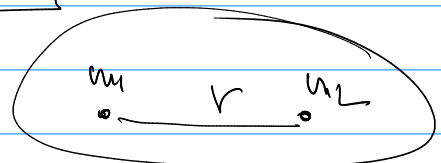
$$\approx \frac{GMm}{(R+h)^2} = mg$$

$$\approx \boxed{g = \frac{GM}{R^2}} \quad R \gg h$$

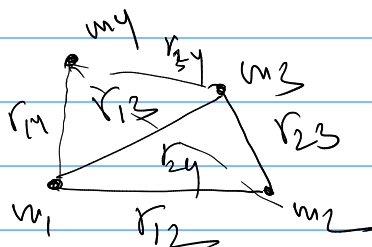
$$(a+x)^n = a^n + na + \dots$$



$$U = -\frac{Gm_1m_2}{r}$$



$$W = U_2 = -\frac{Gm_1m_2}{r}$$

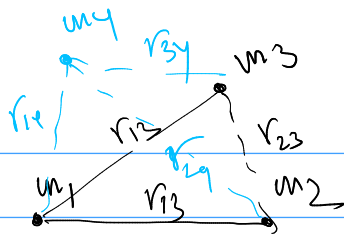


$$U = U_{12} + U_{13} + U_{23}$$

$$U_{12} = -\frac{Gm_1m_2}{r_{12}} \quad U_{13} = -\frac{Gm_1m_3}{r_{13}}$$

$$U_{23} = -\frac{Gm_2m_3}{r_{23}}$$

$$U =$$



$$U_3 = U_{13} + U_{23}$$

$$U_{12} = -\frac{Gm_1m_2}{r_{12}}$$

$$U = U_{12} + U_3$$

$$U = U_{12} + U_{13} + U_{23}$$

$$U_4 = U_{14} + U_{24} + U_{34}$$

$$U = U_{12} + U_4 + U_3$$

$$U = U_{12} + U_{13} + U_{23} + U_{14} + U_{24} + U_{34} + \dots$$

$$U = -\frac{GMm}{r}$$

$$\frac{du}{dr} = +\frac{GMm}{r^2} = F$$

$$F = -\frac{du}{dr}$$

$$\int du = \int F dr$$

$$\Delta U = -\int F dr = -\Delta W$$

$$\Delta W = -\Delta U$$

$$F = -kx$$

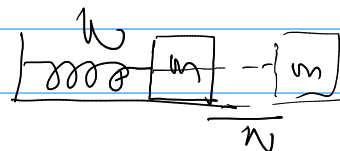
$$F = -\frac{du}{dx}$$

$$F = kx = -\frac{du}{dx}$$

$$\int du = \int kx dx$$

$$u = \frac{1}{2} kx^2$$

$$u = \frac{1}{2} kx^2$$



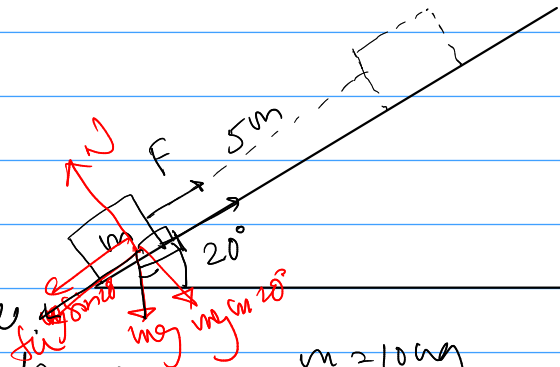
(i) Work done by the gravity.

(ii) How much energy lost due to friction.

(iii) Work done by applied force.

(iv) What is the change of kinetic energy of the block after travelling 5m?

(v) What is the speed of the block after travelling 5m.



$$m = 10 \text{ kg}$$

$$v_i = 15 \text{ m/s}$$

$$F = 150 \text{ N}$$

$$\mu_k = 0.40$$

$$d = 5 \text{ m}$$

Ans: (i) $W_g = \vec{F}_g \cdot \vec{d} = mgd \cos 110^\circ$
 $W_g = mg \sin \theta d \cos 180^\circ$

$$W_g = -167.5 \text{ J}$$

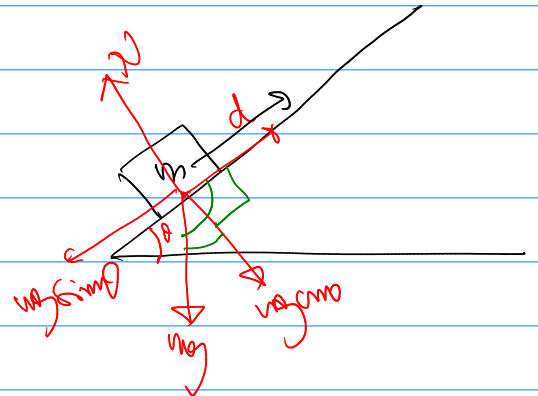
Ans

(ii) $W_{fa} = \vec{f}_a \cdot \vec{d} = f_a d \cos 180^\circ$

$$\Rightarrow W_{fa} = -f_a d = -\mu_k mg d$$

$$\Rightarrow W_{fa} = -184 \text{ J}$$

$$W_{fa} = -184 \text{ J}$$



(iii) $W_a = \vec{F}_a \cdot \vec{d} = F_a d \cos 0^\circ = 100 \times 5 \times \cos 0^\circ = 500 \text{ J}$

Ans

(iv) $W_{net} = \Delta K$

$$\Rightarrow \Delta K = W_{net} = W_a + W_g + W_{fa} + W_{fs}$$

$$\Rightarrow \Delta K = W_a + W_g + W_{fa}$$

$$\Rightarrow \Delta K = 500 - 167.58 - 184$$

$$\Rightarrow \Delta K = 148.5 \text{ J}$$

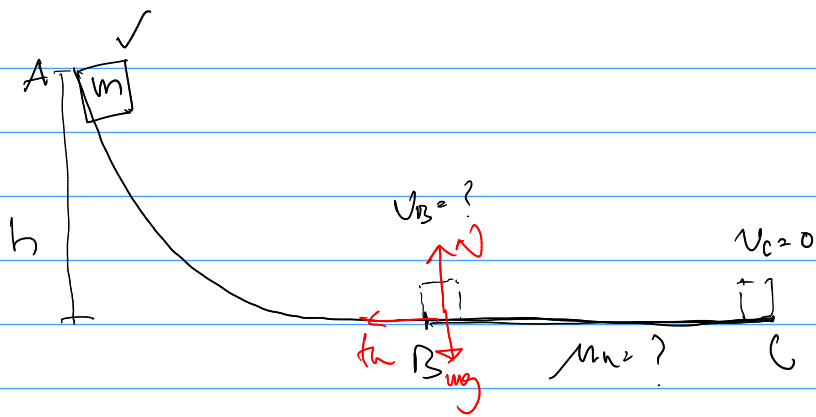
(v) $\Delta K = K_f - K_i = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$

$$\Rightarrow \frac{1}{2} m v_f^2 = \Delta K + \frac{1}{2} m v_i^2$$

$$v_f^2 = \frac{2 \Delta K + m v_i^2}{m}$$

$$v_f = \sqrt{\frac{2 \Delta K + m v_i^2}{m}}$$

Ans



AB = frictionless
BC = frictional

$h = 15 \text{ cm}$
 $BC = 100 \text{ cm}$
 $\mu = ?$

$$W_{\text{net}} = \Delta K = K_C - K_B$$

$$\Rightarrow W_{\text{net}} = -W_B$$

$$\Rightarrow \cancel{W_A} + \cancel{W_g} + W_f = -W_B$$

$$\Rightarrow W_f = -W_B$$

$$\Rightarrow f_k d = f W_B$$

$$\Rightarrow \mu N d = \frac{1}{2} m v_B^2$$

$$\Rightarrow \mu m g d = \frac{1}{2} m v_B^2$$

$$\Rightarrow \mu = \frac{v_B^2}{2gd}$$

$$\Rightarrow \mu = \frac{2gh}{2gd}$$

$$\Rightarrow \boxed{\mu = \frac{h}{d}}$$

$$\boxed{\mu = 0.15} \text{ Ans}$$

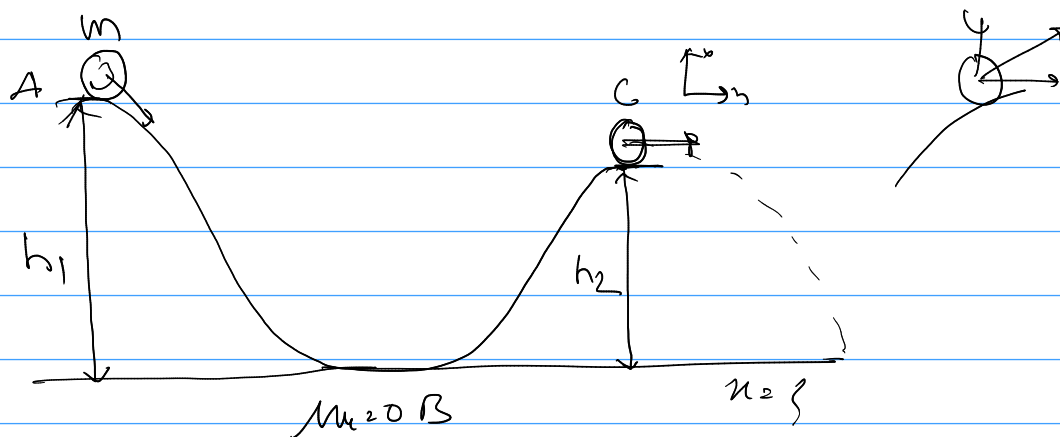
$$E_A = E_B$$

$$\Rightarrow \cancel{K_A} + U_A = K_B + U_B$$

$$\Rightarrow U_A = K_B$$

$$\Rightarrow mgh = \frac{1}{2} m v_B^2$$

$$\Rightarrow v_B = \sqrt{2gh}$$



$$h_1 = 100 \text{ m}$$

$$h_2 = 50 \text{ m}$$

Solⁿ
r

$$x = v_{0x} t$$

$$v_{0x} = v_c$$

$$E_A = E_C$$

$$\frac{1}{2} m v_A^2 + U_A = \frac{1}{2} m v_C^2 + U_C$$

$$m g h_1 = \frac{1}{2} m v_C^2 + m g h_2$$

$$v_C^2 = 2g(h_1 - h_2)$$

$$v_C = \sqrt{2g(h_1 - h_2)}$$

$$v_C = 31.3 \text{ m/s}$$

$$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$$

$$-50 = 0 + 0 - 4.9 t^2$$

$$t = \sqrt{\frac{50}{4.9}} = 3.19 \text{ s}$$

$$x = v_{0x} t = 31.3 \times 3.19 \approx 99.8 \text{ m} \quad \underline{Ans}$$